Process uncertainty

$$N_{t+1} \sim \text{lognormal}\left(\ln(\mu_{t+1}), \sigma_{proc}\right)$$

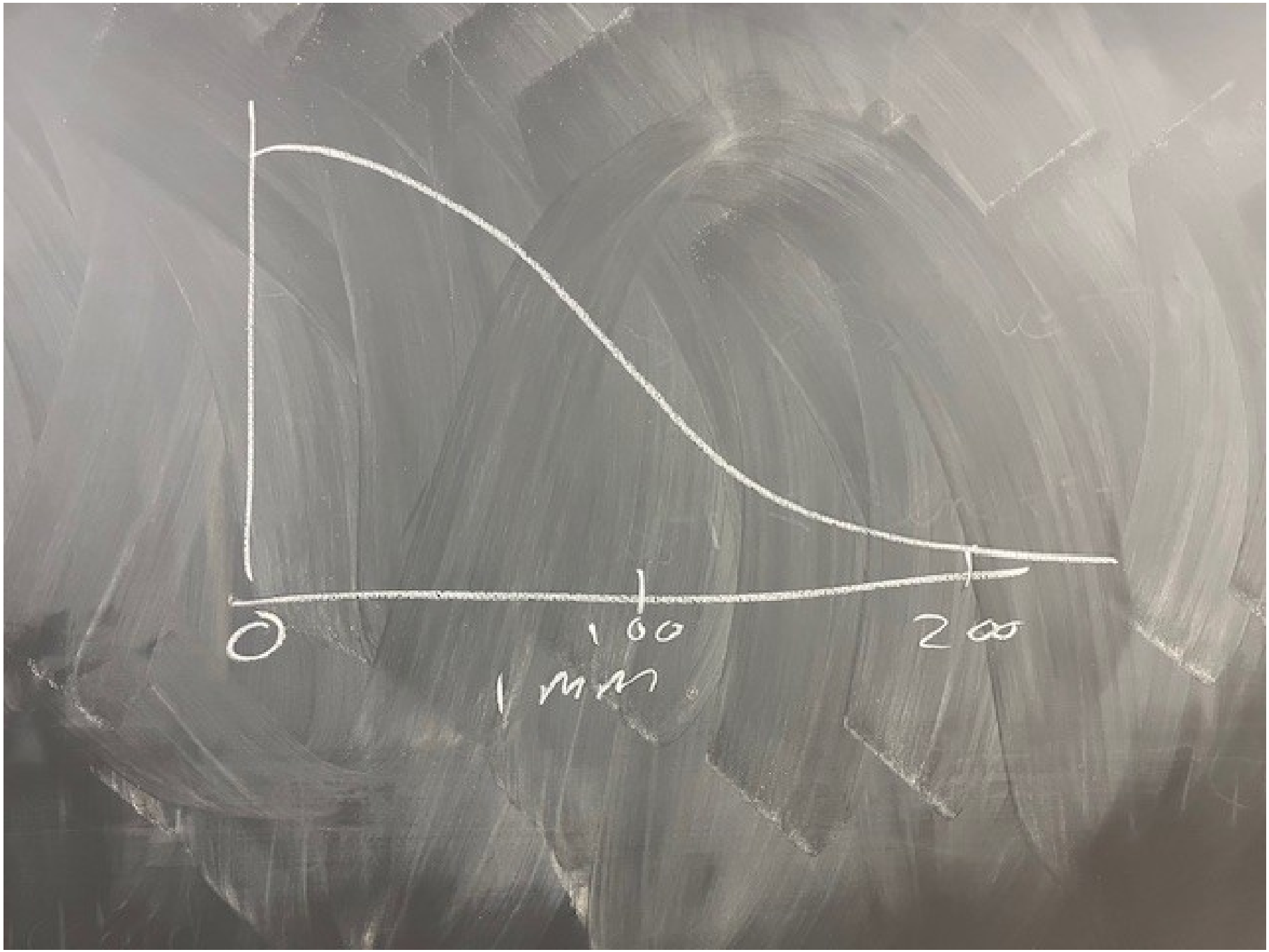
Deterministic skeleton. Or mean of biology

Expectation $\rightarrow \mu_{t+1} = i + s N_t$

$$D_t \sim \text{lognormal}\left(\ln(N_t), \sigma_{obs}\right)$$

$$\ln(N_{t+1}) = \ln(\mu_{t+1}) + \sigma_{proc} Z_t$$

$$Z_t \sim \text{normal}(0, 1)$$


0
100
200
1 MA